[fn39] The gravitational potential.

[fn40] $x(t) = ½\ at^2$.

[fn41] Especially since he had become excited …

[fn42] The actual structure of the vertices of the spinfoam is a bit more complex than the one in figure 7.2, and resembles more closely the one shown in figure 7.4.

[fn43] It's a Feynman diagram because it is a history of quanta, as in the Feynman diagrams. Except that now, the quanta are not quanta moving in space, but rather quanta *of* space. The graph they draw in their interactions is not a representation of the movement of particles in space, but represents the plot of space itself. But the resulting picture is also precisely a lattice like the one used in the lattice approximation, because it represents a discretized spacetime. With the difference that it is no longer an approximation, but the *real* discrete structure of space at a small scale.

[fn44] The first defines the Hilbert space of the theory. The second describes the algebra of the operators. The third describes the size of transition of each vertex, such as the one shown in figure 7.4.

[fn45] '[…] all the different elementary particles could be reduced to one universal substance which could equally be called energy or matter, and none of the particles should be privileged and considered more fundamental. This point of view corresponds to Anaximander's doctrine, and I am convinced that in modern physics this is the correct point of view.' Werner Heisenberg, *Physics and Philosophy: The Revolution in Modern Science* (New York, Harper & Row, 1962).